Hypothesis Testing with E-Values

Chapter 7: Sequential Anytime-Valid Inference using E-Processes

Based on Ramdas & Wang

Outline

Notation and background you need I

This chapter is entirely about *sequential* data: $X_1, X_2, X_3, \ldots$ arriving one at a time. Everything hinges on four ideas that are new if you have only seen fixed-sample-size statistics. We build each one from scratch with a plain-English analogy first, then the formal definition.

(a) Filtration — your growing folder of evidence.
Intuition: Imagine a folder that only ever gains pages. Page t contains everything you knew right after seeing the t-th data point. You can look back at old pages any time, but you can never un-see what is already inside, and you cannot peek at future pages. A *filtration* $(\mathcal{F}_t)_{t\in\mathbb{T}}$ is the mathematical version of this folder: a nested, non-decreasing sequence of σ-algebras,

$$\mathcal{F}_0 \subseteq \mathcal{F}_1 \subseteq \mathcal{F}_2 \subseteq \cdots ,$$

where $\mathcal{F}_t$ represents "all events you can decide are true or false using only data up to time t." Usually $\mathcal{F}_t = \sigma(X_1, \ldots, X_t)$, the *natural filtration* generated by the data itself, with $\mathcal{F}_0 = \{\varnothing, \Omega\}$ (trivial — you know nothing yet). A process $Y = (Y_t)$ is *adapted* to $\mathcal{F}$ if Y_t is measurable w.r.t. $\mathcal{F}_t$ for every t — informally, Y_t is computable from data available by time t, no peeking ahead allowed. A process is *predictable* if Y_t is measurable w.r.t. $\mathcal{F}_{t-1}$ — it must be decided *before* seeing X_t.

Notation and background you need II

(b) Stopping time — a rule for when to stop looking.
Intuition: You are flipping a coin and you decide: "I'll stop the very first time I see 3 heads in a row." This is a legitimate rule because, at every moment, you can tell (using only what you've seen so far) whether to stop right now. Contrast this with an illegitimate rule like "I'll stop one flip before the 10th head" – you cannot know in advance which flip that is without seeing the future.
A *stopping time* τ is a random variable taking values in $\{0, 1, 2, \dots\} \cup \{\infty\}$ such that

$$\{\tau \leq t\} \in \mathcal{F}_t \quad \text{for every } t.$$

In words: at every time t, you can decide (using only $\mathcal{F}_t$, i.e., data so far) whether the rule has told you to stop by now. Stopping times formalize "peeking at the data and deciding to stop" — as long as the decision to stop at time t only uses information available at time t.

Notation and background you need III

(c) Supermartingale — wealth in a game rigged against you (or fair, at best).
Intuition: Think of M_t as your wealth after t rounds of a betting game. A *supermartingale* means: given everything you know so far, your expected wealth tomorrow is at most your wealth today. The game may be unfavorable or exactly fair, but never favorable, on average, one step ahead.
Formally, $M = (M_t)$ is a supermartingale for $\mathbb{P}$ if it is adapted, integrable, and

Supermartingale

$$\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] \leq M_{t-1} \quad \text{for all } t \geq 1.$$

If "$\leq$" is replaced by "$=$", M is a *martingale* (a perfectly fair game). A *test (super)martingale* for a set of nulls $\mathcal{P}$ additionally requires: $M_t \geq 0$ a.s. for every $\mathbb{P} \in \mathcal{P}$, and $\mathbb{E}^{\mathbb{P}}[M_0] \leq 1$.

Notation and background you need IV

(d) E-process — your wealth betting against the null.
Intuition: Start with \$1. At every step you place a bet against the null hypothesis being true. If the null is true, your bet is designed so you cannot expect to profit (on average, wealth does not grow); if the null is false, a good strategy lets your wealth grow, potentially to enormous multiples of your starting \$1. The path of your wealth over time is the *e-process*. Large wealth at any point you choose to stop and look is evidence against the null.

E-process (Definition 7.3)

A sequence $E = (E_t)_{t \in \mathbb{T}}$ adapted to $\mathcal{F}$ is an e-process for $\mathcal{P}$ if either (equivalent) condition holds:

(a) $\mathbb{E}^{\mathbb{P}}[E_\tau] \leq 1$ for every stopping time τ and every $\mathbb{P} \in \mathcal{P}$;

(b) there is a test martingale family $(M^{\mathbb{P}})_{\mathbb{P} \in \mathcal{P}}$ with $E \leq M^{\mathbb{P}}$ a.s., for each $\mathbb{P} \in \mathcal{P}$.

Every test (super)martingale is an e-process, but not every e-process is a (super)martingale — e-processes form a strictly larger, more flexible class. This chapter's entire agenda is: build such wealth processes, and show they give valid sequential tests *no matter when or why you decide to stop*.

Notation and background you need V

Why this matters, in one sentence. A fixed-sample-size test (e.g., a classical p-value at a pre-announced n) is only guaranteed valid if you look at the data exactly once, at the n you fixed in advance. An e-process is a wealth process that stays a valid measure of evidence *no matter which (random, data-dependent) time* τ *you choose to stop and look* — this is what "anytime-valid" means, and it is the whole point of sequential anytime-valid inference (SAVI).

7.1 Peeking at p-values: the problem I

Intuition. A scientist collects data one subject at a time and, being impatient (or economical), checks the p-value after every few subjects. If it ever dips below 0.05, she stops and declares victory. This feels harmless — after all, each individual p-value is a legitimate 5%-level test. But looking at *many* of them and stopping at the first lucky dip is a completely different, and much riskier, procedure.

Setup. Suppose $P_1, P_2, \ldots$ is a sequence of p-values computed after $1, 2, 3, \ldots$ subjects (assume for now each P_n is exactly Uniform$(0,1)$ under the null, as classical theory promises). Define the stopping time

$$\tau := \inf\{n \geq 1 : P_n \leq \alpha\}.$$

The heart of the issue

Even though, for every *fixed* n,

$$\mathbb{P}(P_n \leq \alpha) \leq \alpha,$$

it is typically the case (e.g., for a t-test p-value) that $\tau < \infty$ almost surely under the null, and

$$\mathbb{P}(P_\tau \leq \alpha) = 1.$$

7.1 Peeking at p-values: the problem II

That is: if you are willing to wait long enough, you are *guaranteed* to see a "significant" result, even when the null is exactly true! This is called **sampling to a foregone conclusion**.

7.1 Peeking at p-values: the problem III

Why does this happen? A valid p-value guarantee, $\mathbb{P}(P_n \leq \alpha) \leq \alpha$, is a statement about one fixed, pre-chosen n. It says nothing about the probability that *some* P_n along the way dips below α. Under the null, P_n fluctuates forever (it does not converge to anything sensible), and a fluctuating sequence started arbitrarily many times will eventually hit any fixed threshold with probability 1 (much like a symmetric random walk eventually crosses any level).

Consequences.

- This is believed to be a real contributor to the "reproducibility crisis" in psychology, biomedicine, and social science: researchers who informally check their data as it accumulates (even with good intentions) can inflate the true type-I error far above the nominal α.
- The problem cannot be diagnosed from the data alone — you would need to know the exact protocol (did they peek? how often?) to correct for it, and this information is rarely reported.
- Rather than trying to *ban* peeking (unenforceable), this chapter's approach is to design procedures that remain valid *no matter how or when* you peek. That is exactly what e-processes deliver.

7.2 Wald's SPRT: a motivating numerical example I

Setup. Data $X_1, X_2, \ldots$ are iid Bernoulli(p). Null: $p = 1/2$. Alternative: $p = 0.6$. Let f_0, f_1 be the null/alternative densities. The likelihood ratio process is

Likelihood ratio

$$L_n = \prod_{i=1}^{n} \frac{f_1(X_i)}{f_0(X_i)}.$$

L_n is an e-variable at each fixed n; the sequence $(L_t)_{t\in\mathbb{N}}$ is in fact an e-process.

Fixed-sample Neyman–Pearson LRT. For target type-I error $\alpha = 0.05$ and type-II error $\beta = 0.05$, a calculation shows $n = 280$ and the optimal rule rejects when $L_{280} \geq \gamma = 0.962$ (equivalently, at least 154 heads out of 280). Curiously, $\gamma < 1$: we reject even though the alternative's likelihood is only marginally larger than the null's.

Wald's SPRT. Instead of waiting for a fixed n, update L_t after every observation:

- if $L_t \geq \gamma_1 = 1/\alpha$: reject the null and stop;
- if $L_t \leq \gamma_0 = \beta$: accept (fail to reject) and stop;

7.2 Wald's SPRT: a motivating numerical example II

- otherwise: keep sampling.

These conservative thresholds $\gamma_0 = \beta,\ \gamma_1 = 1/\alpha$ keep type-I and type-II errors below α, β respectively, as a consequence of Ville's inequality (Section 7.4). For our numbers: reject if $L_t \geq 20$, accept if $L_t \leq 0.05$.

7.2 Wald's SPRT: a motivating numerical example III

Data-generating dist. Setting	$p = 0.5$ null	$p = 0.55$ misspecified	$p = 0.6$ alternative	$p = 0.65$ misspecified
SPRT expected sample size	137.7	234.5	139.3	76.1
LRT pre-specified sample size	280	280	280	280
Power of SPRT	0.04	0.52	0.96	1.00
Power of LRT	0.05	0.48	0.95	1.00

Table: Comparison of SPRT vs. fixed-sample LRT (10,000 Monte Carlo repetitions). Table 7.1 of the text.

Punchline. Under both the null and the alternative, the SPRT stops, on average, using roughly *half* the samples of the fixed-design LRT (137.7 or 139.3 vs. 280), while achieving comparable power. Sequential testing is not just "the same test done incrementally" — it can be dramatically more efficient.

Theorem 7.2: Wald's SPRT optimality (informal)

Let α, β be LRT errors with n samples; let α', β' be SPRT errors with stopping time τ. If $\alpha \leq \alpha'$ and $\beta \leq \beta'$, then $\max(\mathbb{E}^{\mathbb{P}}[\tau], \mathbb{E}^{\mathbb{Q}}[\tau]) \leq n$: no other rule with equal or better error control needs a smaller expected sample size than the SPRT.

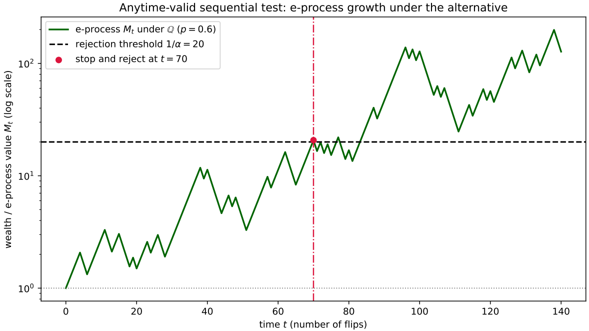
7.2 Wald's SPRT: a motivating numerical example IV

How SAVI differs from Wald's classical paradigm (four points):

1. **Decisions vs. evidence:** Wald wants a binary accept/reject decision with pre-set errors; SAVI wants a continuously-updatable *measure of evidence* (the e-process) — decisions are a secondary, optional use.
2. **One stopping rule vs. all stopping rules:** Wald designs one specific stopping rule in advance; an e-process is valid *simultaneously at every possible stopping time*, planned or not.
3. **Optimality criteria differ:** Wald optimizes the error-vs.-sample-size tradeoff; SAVI's "log-optimality" (introduced below) optimizes the growth rate of evidence. For simple null/alternative these happen to coincide, but not in general.
4. **Beyond likelihood ratios:** Wald's classical theory needs likelihood ratios to exist (parametric, common reference measure). E-processes give a universal language that works for arbitrary composite, even nonparametric, nulls.

Visualizing anytime-valid sequential testing under the alternative

Setup. Bet on iid Bernoulli data using the likelihood-ratio strategy towards $q = 0.6$ (factor 1.2 on heads, 0.8 on tails), but now the *true* bias is $p = 0.6$ (the alternative is correct). The e-process should grow exponentially on average, and we may stop and reject as soon as it crosses the rejection threshold $1/\alpha$ — we do not need to wait for a pre-specified sample size.



The path fluctuates (bets can lose in the short run) but trends upward; here it first crosses the threshold $1/\alpha = 20$ at $t = 70$, at which point we may stop and reject H_0 — exactly the anytime-valid analogue of Wald's SPRT rejection rule from Section 7.2, but now understood as a special case of thresholding a

7.3 Four SAVI tools

Intuition. There are four closely related objects for sequential inference; two are tied to a fixed significance level α and two are not.

- **e-processes** (no fixed α) — our main object: sequential analog of e-variables.
- **p-processes** (no fixed α) — sequential analog of p-variables.
- **level-α sequential tests** — sequential analog of level-α tests.
- $(1-\alpha)$-**confidence sequences** — sequential analog of confidence intervals.

We focus on e-processes: the others follow from Ville's inequality (next section) applied to one or more e-processes.

7.3 Level-α sequential tests

Intuition. A sequential test is a rule that, at each time t, outputs a binary "reject / don't reject" decision $\phi_t \in \{0, 1\}$ using only data so far, and once it rejects, it stays rejected. Validity means: the chance you *ever* falsely reject (across all of time) is at most α.

Level-α sequential test

A binary adapted process $\phi = (\phi_t)$ is a level-α sequential test for $\mathcal{P}$ if

$$\sup_{\mathbb{P} \in \mathcal{P}} \mathbb{P}(\exists t \geq 1 : \phi_t = 1) \leq \alpha. \tag{7.1}$$

Equivalently, letting $\tau := \inf\{t \geq 1 : \phi_t = 1\}$ (a stopping time; $\inf \varnothing = \infty$), the condition becomes

$$\mathbb{E}^{\mathbb{P}}[\phi_\tau] \leq \alpha \quad \text{for all } \mathbb{P} \in \mathcal{P}, \; \tau \in \mathcal{T}. \tag{7.2}$$

Here ϕ_t: reject-or-not indicator at time t; $\mathcal{T}$: set of all $\mathcal{F}$-stopping times.

7.3 Test (super)martingales and e-processes: formal definitions

Definition 7.3

(i) $M = (M_t)_{t \in \mathbb{T}}$ is a *test (super)martingale* for $\mathcal{P}$ if for every $\mathbb{P} \in \mathcal{P}$: (a) $M \geq 0$ a.s.; (b) M is a (super)martingale under $\mathbb{P}$; (c) $\mathbb{E}^{\mathbb{P}}[M_0] \leq 1$.
(ii) An e-process $E = (E_t)$ adapted to $\mathcal{F}$ satisfies either (equivalent):

(a) $\mathbb{E}^{\mathbb{P}}[E_\tau] \leq 1$ for any stopping time $\tau \in \mathcal{T}$, any $\mathbb{P} \in \mathcal{P}$;

(b) $\exists$ a test martingale family $(M^{\mathbb{P}})_{\mathbb{P} \in \mathcal{P}}$ with $E \leq M^{\mathbb{P}}$ a.s., for each $\mathbb{P}$.

The equivalence of (a) and (b) is nontrivial (uses Doob's decomposition); we take it as given. Every test (super)martingale is trivially an e-process (take $M^{\mathbb{P}} = M$), but the converse fails: e-processes are a strictly larger, more flexible class.

Definition 7.4: asymptotic growth rate

An e-process M is *consistent* against $\mathbb{Q}$ if $M_t \to \infty$, $\mathbb{Q}$-a.s. The (typically $\mathbb{Q}$-a.s. constant) quantity $\liminf_{t \to \infty} \frac{\log M_t}{t}$ is the *asymptotic growth rate* of M against $\mathbb{Q}$. It is always ≤ 0 against $\mathbb{P} \in \mathcal{P}$.

Python: simulating a supermartingale

We simulate a simple test martingale directly from Definition 7.3: bet on iid fair coin flips ($X_i \in \{0, 1\}$) using the likelihood-ratio bet towards $q = 0.6$, i.e., multiply wealth by 1.2 on heads and 0.8 on tails. Under the true null ($p = 0.5$), this wealth process is a genuine *martingale* (hence a supermartingale): $\mathbb{E}[\text{next factor}] = 0.5(1.2) + 0.5(0.8) = 1$.

```
import numpy as np

def simulate_wealth(n_steps, p_true, q=0.6, seed=0):
    rng = np.random.default_rng(seed)
    heads = rng.random(n_steps) < p_true
    factor = np.where(heads, 2*q, 2*(1-q))   # 1.2 if heads, 0.8 if tails
    wealth = np.concatenate(([1.0], np.cumprod(factor)))
    return wealth

# Under H0 (p=0.5): E[factor] = 0.5*1.2 + 0.5*0.8 = 1.0  -> martingale
w_null = simulate_wealth(400, p_true=0.5, seed=1)
print("wealth at t=400 under H0:", round(w_null[-1], 4))

# Check the martingale property empirically: average factor over many paths
factors = []
for s in range(20000):
    heads = np.random.default_rng(s).random(1) < 0.5
    factors.append(1.2 if heads[0] else 0.8)
print("empirical E[factor | H0] =", round(np.mean(factors), 4), " (theory: 1.0)")
```

7.4 The optional stopping theorem I

Intuition. If wealth M never grows in expectation one step ahead (a supermartingale), then no matter which stopping time τ you pick to cash out — even a rule that peeks at the whole history and cleverly decides when to stop — your expected wealth at that random time is still at most where you started. You cannot out-smart a fair (or unfavorable) game by choosing *when* to stop.
Supermartingale convergence. Any test supermartingale M for $\mathbb{P}$ has a well-defined (finite) $\mathbb{P}$-almost sure limit $M_\infty := \lim_n M_n$.

Fact 7.5: Optional stopping theorem

If M is a test (super)martingale for $\mathbb{P}$, then for any stopping times $\sigma \leq \tau$,

$$\mathbb{E}^{\mathbb{P}}[M_\tau \mid \mathcal{F}_\sigma] \leq M_\sigma.$$

Taking $\sigma = 0$: $\mathbb{E}^{\mathbb{P}}[M_\tau] \leq 1$. Consequently, if M is an e-process for $\mathcal{P}$, then $\mathbb{E}^{\mathbb{P}}[M_\tau] \leq 1$ for all $\mathbb{P} \in \mathcal{P}$.

Notably, this version holds for *any* stopping time τ, with no boundedness restriction — a consequence of the nonnegativity of M.

7.4 The optional stopping theorem II

Intuition for Ville's inequality. Combine optional stopping with Markov's inequality: since $\mathbb{E}^{\mathbb{P}}[M_\tau] \leq 1$ for *every* stopping time τ (not just fixed times), we can bound the probability that the wealth process *ever* gets as large as $1/\alpha$ — not just at one time, but over its entire trajectory.

Fact 7.6: Ville's inequality

If M is an e-process for $\mathcal{P}$, the following equivalent statements hold for every $\alpha \in [0, 1]$:

$$\mathbb{P}\left(\exists t \in \mathbb{N} : M_t \geq \tfrac{1}{\alpha}\right) \leq \alpha \quad \text{for every } \mathbb{P} \in \mathcal{P}; \tag{Ville}$$

$$\iff \sup_{\mathbb{P}\in\mathcal{P}} \mathbb{P}\left(\sup_{t\in\mathbb{N}} M_t \geq \tfrac{1}{\alpha}\right) \leq \alpha; \tag{7.4a}$$

$$\iff \sup_{\mathbb{P}\in\mathcal{P},\, \tau\geq 0} \mathbb{P}\left(M_\tau \geq \tfrac{1}{\alpha}\right) \leq \alpha, \quad \tau \text{ any stopping time.} \tag{7.4b}$$

7.4 The optional stopping theorem III

Step-by-step proof idea of Ville's inequality.
Step 1 (define the crossing time). Fix $\alpha \in (0, 1]$ and let

$$\tau^* := \inf\{t \geq 0 : M_t \geq 1/\alpha\}$$

(with $\inf \varnothing = \infty$). This is a valid stopping time: whether $\tau^* \leq t$ depends only on $M_0, \ldots, M_t$, all $\mathcal{F}_t$-measurable.
Step 2 (apply optional stopping at τ^*). By Fact 7.5 (optional stopping), for every finite n,

$$\mathbb{E}^{\mathbb{P}}[M_{\tau^* \wedge n}] \leq \mathbb{E}^{\mathbb{P}}[M_0] \leq 1,$$

since $M_{\tau^* \wedge n}$ is the wealth process stopped at the (finite, bounded) time $\tau^* \wedge n$, and stopped supermartingales are still supermartingales.
Step 3 (Markov's inequality on the event of crossing). On the event $\{\tau^* \leq n\}$, we have $M_{\tau^* \wedge n} = M_{\tau^*} \geq 1/\alpha$ by definition of τ^* (assuming no "jump over" the threshold, which holds since we check every integer time). Hence

$$\mathbb{P}(\tau^* \leq n) \leq \alpha \, \mathbb{E}^{\mathbb{P}}[M_{\tau^* \wedge n}] \leq \alpha.$$

7.4 The optional stopping theorem IV

Step 4 (let $n \to \infty$). $\{\tau^* \le n\} \uparrow \{\tau^* < \infty\} = \{\exists t : M_t \ge 1/\alpha\}$, so by continuity of probability,

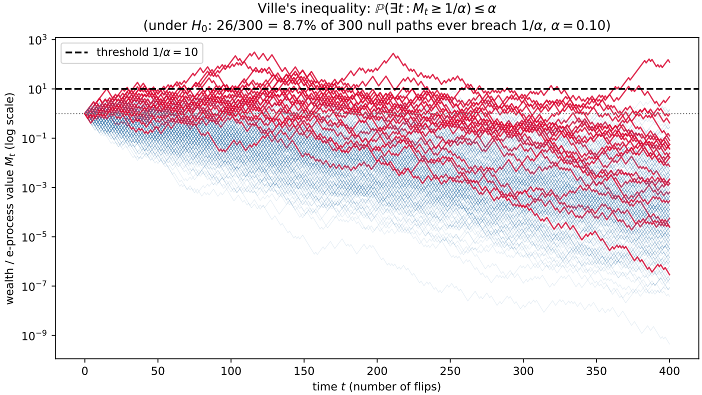
$$\mathbb{P}(\exists t : M_t \ge 1/\alpha) = \lim_{n\to\infty} \mathbb{P}(\tau^* \le n) \le \alpha. \qquad \blacksquare$$

7.4 The optional stopping theorem V

Reading the inequality. For any *fixed* t, M_t is itself an e-variable, so $\mathbb{P}(M_t \geq 1/\alpha) \leq \alpha$ is just ordinary Markov's inequality. Ville's inequality is the *time-uniform* generalization: the bound holds simultaneously across the entire (possibly infinite) trajectory, not just at one time slice. This is exactly what makes stopping at a data-dependent time safe.

Practical meaning. Defining $\phi_t := \mathbb{1}\{M_t \geq 1/\alpha\}$ turns M into a level-α sequential test (Fact 7.6 $\Rightarrow$ Definition of level-α test). The picture: most null trajectories wander near or below 1 and never cross $1/\alpha$; only an α-fraction ever breach it.

7.4 The optional stopping theorem VI



Proposition 7.8: the converse

Every level-α sequential test ϕ for $\mathcal{P}$ can be obtained by thresholding some e-process M for $\mathcal{P}$ at $1/\alpha$.

Proof. Define $M_t := \phi_t/\alpha$. Then M_t only takes values 0 or $1/\alpha$, so $M_t \geq 1/\alpha \iff \phi_t = 1$. It remains to check M is an e-process: for any stopping time τ and any $\mathbb{P} \in \mathcal{P}$,

$$\mathbb{E}^{\mathbb{P}}[M_\tau] = \mathbb{E}^{\mathbb{P}}[\phi_\tau]/\alpha \leq 1$$

by (7.2). ■

Conclusion: e-processes and level-α sequential tests are two sides of the same coin — thresholding one direction gives a test, and every test arises this way. But e-processes carry *more* information (the actual wealth/evidence value) than a binary reject/don't-reject decision, and remain interpretable without ever thresholding.

7.4 Optional continuation and p-processes

Intuition (optional continuation). Suppose scientist A runs an experiment and produces e-process M^A up to some point (possibly a random, data-dependent point τ). Scientist B can then take over, starting a fresh e-process M^B (with $M_t^B = 1$ while $t \leq \tau$), and *multiply* the two together. The combined wealth is still a valid e-process — evidence can be handed off between experimenters, or across looks, without breaking validity.

Proposition 7.9

Let M^A, M^B be e-processes for $\mathcal{P}$ and τ a stopping time with $M_t^B = 1$ on $\{t \leq \tau\}$. Define $M_t := M_{\tau \wedge t}^A M_t^B$. Then M is an e-process for $\mathcal{P}$.

This means: switching betting strategies (even multiple times, even using human judgment about when to switch) never invalidates the e-process, as long as each strategy is itself valid on its own turn.

Definition 7.10: p-process

A nonnegative process $P = (P_t)$ is a *p-process* for $\mathcal{P}$ if P_τ is a p-variable for every stopping time τ and every $\mathbb{P} \in \mathcal{P}$.

Ville's inequality (7.4a) immediately implies: if M is an e-process, then $\left(\inf_{s \leq t} 1/M_s\right)_t$ is a p-process.
Intuition: an e-process reports evidence *now*; a p-process reports the *best* evidence ever seen so far —

7.5.1 Log-optimality of the likelihood ratio process

Intuition. If testing a simple $\mathbb{P}$ against a simple $\mathbb{Q}$, the natural e-process is just the running likelihood ratio — exactly the object from the SPRT. It turns out this is not just *a* valid e-process, it is the *best possible* one, in a precise sense (log-optimality).

Likelihood ratio process

$$M_0 = 1, \qquad M_t = \prod_{i=1}^{t} \frac{\mathrm{d}\mathbb{Q}}{\mathrm{d}\mathbb{P}}(X_i) \quad \text{for } t \in \mathbb{N}. \tag{7.5}$$

M is not just an e-process, but a genuine *test martingale* for $\mathbb{P}$ (even without the iid assumption, using the running likelihood ratio of the first n points).

Theorem 7.11: log-optimality

For any stopping time τ finite $\mathbb{Q}$-a.s., and any e-process M' for $\mathbb{P}$,

$$\mathbb{E}^{\mathbb{Q}}[\log M_\tau] \geq \mathbb{E}^{\mathbb{Q}}[\log M'_\tau]. \tag{7.7}$$

Proof idea: M'_τ is measurable w.r.t. $\mathcal{F}_\tau$; for any $\mathbb{P}$-null event $A \in \mathcal{F}_\tau$, one shows $\mathbb{Q}(A) = 0$ too (using $\mathbb{E}^{\mathbb{P}}[\mathbb{1}_A \mathbb{1}_{\{\tau \leq n\}} M_n] = 0$ and taking $n \to \infty$), so $\mathbb{Q} \ll \mathbb{P}$ on $\mathcal{F}_\tau$; a general e-variable optimality result (Proposition 3.22) then finishes the proof. No other e-process can beat the likelihood ratio's expected

7.5.2 Asymptotic log-optimality and the plug-in method I

Intuition. With iid data, the running likelihood ratio's growth rate settles down to a constant: the Kullback–Leibler divergence $\mathrm{KL}(\mathbb{Q}, \mathbb{P})$, by the law of large numbers. No e-process can grow faster asymptotically.

Proposition 7.12

In the iid setting, the likelihood ratio process has asymptotic growth rate $\mathrm{KL}(\mathbb{Q}, \mathbb{P}) > 0$ (hence is consistent); no other e-process has a larger asymptotic growth rate.

When $\mathbb{Q}$ is composite (unknown alternative): plug in an estimate $\hat{\theta}_i$ of the alternative parameter, updated online:

$$M_0 = 1, \qquad M_t = \prod_{i=1}^{t} \ell(X_i; \theta_i), \quad \theta_i \text{ measurable w.r.t. } \sigma(X_1, \ldots, X_{i-1}). \tag{7.8}$$

7.5.2 Asymptotic log-optimality and the plug-in method II

Definition 7.13: asymptotic log-optimality

M is asymptotically log-optimal for $\mathbb{P}$ against $\mathcal{Q}$ if for every $\mathbb{Q} \in \mathcal{Q}$,

$$\lim_{t\to\infty} \frac{1}{t}\big(\log M_t - \log M_t^{\mathbb{Q}}\big) \geq 0 \quad \text{in } L^1(\mathbb{Q}),$$

where $M^{\mathbb{Q}}$ is the (oracle) likelihood ratio process for $\mathbb{Q}$.

Proposition 7.14 (sketch) and Corollary 7.15

If $\log \ell(x;\theta)$ is concave in θ with gradient $\eta(x;\theta)$, and $\theta_i \to \theta$ in $L^2(\mathbb{Q}_\theta)$ with $\sup_i \mathbb{E}^{\mathbb{Q}_\theta}[\|\eta(X_i;\theta_i)\|^2] < \infty$, then M in (7.8) is asymptotically log-optimal. *Proof uses concavity to lower-bound the log-ratio gap by an inner product, then Cauchy–Schwarz to show it vanishes.* E.g., for $\mathbb{P} = N(\theta_0, 1)$ vs. $\mathbb{Q}_\theta = N(\theta, 1)$, using $\theta_i =$ running sample mean (the MLE) satisfies these conditions since $\mathbb{E}^{\mathbb{Q}_\theta}[(\theta_i - \theta)^2] = 1/(i-1) \to 0$.

7.6 Testing by betting: the game

Intuition. Recast the whole construction as a literal gambling game. The statistician ("Skeptic") bets on each new observation *before* seeing it. Two requirements make this a valid test:

(a) if the null is true, no strategy can make money *systematically* (only by luck);

(b) if the null is false, some strategy *can* guarantee long-run profit.

Final wealth (relative to the $1 start) is then a direct, interpretable measure of evidence against the null.

Algorithm 7.16: Testing by betting (simple null $\mathbb{P}$)

1. Initial wealth $W_0 = 1$.
2. For $t = 1, 2, \ldots$: choose bet $S_t : \mathcal{X} \to [0, \infty]$ with $\mathbb{E}^{\mathbb{P}}[S_t(X) \mid X_1, \ldots, X_{t-1}] \le 1$.
3. Observe X_t; update $W_t = W_{t-1} \cdot S_t(X_t)$.

The constraint on S_t is exactly "S_t is a conditional e-variable for $\mathbb{P}$." This immediately makes W a test supermartingale for $\mathbb{P}$, for *any* betting strategy satisfying the constraint.
Optimal bet. If the alternative $\mathbb{Q}$ is simple and $\mathbb{P} \ll \mathbb{Q}$, the exponent-optimal bet (maximizing $\mathbb{E}^{\mathbb{Q}}[\log S_t]$ subject to the constraint) is exactly the conditional likelihood ratio of $\mathbb{Q}$ to $\mathbb{P}$ — recovering (7.5).

7.6 Testing by betting: composite nulls

Intuition. If the null is composite ($\mathcal{P}$ contains many candidate distributions), the statistician must play a *separate* game against every $\mathbb{P} \in \mathcal{P}$ simultaneously, and report the *worst-case* (most conservative) wealth across all these parallel games as the overall evidence.

Algorithm 7.17: Testing by betting (composite null $\mathcal{P}$)

1. For each $\mathbb{P} \in \mathcal{P}$: initial wealth $W_0^{\mathbb{P}} = 1$.
2. For $t = 1, 2, \ldots$: for each $\mathbb{P}$, choose $S_t^{\mathbb{P}}$ with $\mathbb{E}^{\mathbb{P}}[S_t^{\mathbb{P}}(X) \mid X_1, \ldots, X_{t-1}] \leq 1$.
3. Observe X_t; update $W_t^{\mathbb{P}} = W_{t-1}^{\mathbb{P}} \cdot S_t^{\mathbb{P}}(X_t)$ for each $\mathbb{P}$.
4. Overall wealth: $W_t = \inf_{\mathbb{P} \in \mathcal{P}} W_t^{\mathbb{P}}$.

This is only an *interpretation* — we never literally instantiate uncountably many parallel games in practice; sensible e-processes for composite nulls are usually built more directly (Sections 7.7–7.9). But the betting picture is what makes e-processes intuitive as "evidence," and clarifies why e-processes generalize test supermartingales.

Running example: sequentially testing a coin's fairness I

Setup. $H_0 : \theta = 0.5$ (fair coin). We bet using the fixed likelihood-ratio strategy towards $q = 0.6$: multiply current wealth by 1.2 on heads, by 0.8 on tails. We generate an actual sequence of 30 coin flips (seeded pseudo-randomly, true bias $p = 0.6$) and track wealth W_t after each flip.
The 30 simulated flips (1 $=$ heads, 0 $=$ tails; 19 heads observed):

$$0, 1, 1, 1, 0, 0, 0, 1, 1, 1, 1, 1, 1, 1, 0, 1, 1, 1, 0, 1, 0, 0, 1, 1, 0, 1, 1, 1, 0, 0$$

Running example: sequentially testing a coin's fairness II

t	flip	W_t	t	flip	W_t	t	flip	W_t
1	T	0.800	11	H	1.468	21	T	2.693
2	H	0.960	12	H	1.761	22	T	2.154
3	H	1.152	13	H	2.113	23	H	2.585
4	H	1.382	14	H	2.536	24	H	3.102
5	T	1.106	15	T	2.029	25	T	2.481
6	T	0.885	16	H	2.435	26	H	2.978
7	T	0.708	17	H	2.922	27	H	3.573
8	H	0.849	18	H	3.506	28	H	4.288
9	H	1.019	19	T	2.805	29	T	3.430
10	H	1.223	20	H	3.366	30	T	2.744

Table: Wealth W_t after each of 30 simulated flips ($1.2\times$ on heads, $0.8\times$ on tails).

Running example: sequentially testing a coin's fairness III

What if we stop early at flip 12 vs. continue to flip 30?

- At $t = 12$: wealth $W_{12} = 1.761$. If we report evidence here and stop, this is a perfectly valid e-value: $\mathbb{E}^{\mathbb{P}}[W_{12}] \leq 1$ under H_0, by construction.
- If instead we continue to $t = 30$: wealth $W_{30} = 2.744$. This is *also* a perfectly valid e-value at its own stopping time.
- **Key point:** both are valid *simultaneously*, for *any* stopping rule we might have used to decide between them (even a rule based on how the wealth looked at $t = 12!$). This is exactly Ville's inequality / optional stopping at work: $\mathbb{P}(\exists t \leq 30 : W_t \geq 1/\alpha) \leq \alpha$ regardless of which t we end up choosing.
- Notice wealth is *not* monotonic — it dips (e.g., $t = 5$–7, down to 0.708) before recovering. A classical p-value process peeked at those dips would look "worse than random," but the e-process framework does not care: only the value at the (possibly random) time you actually stop matters, and the guarantee holds regardless.

Running example: sequentially testing a coin's fairness IV

- At $\alpha = 0.05$ (threshold $1/\alpha = 20$), we never come close to rejecting with this particular random sequence — consistent with H_0 being em true and $q = 0.6 \neq \theta_1 = 0.5$ giving only modest average growth per step under H_0 in expectation (growth here is due to the true bias being $p = 0.6$, matching our bet).

Python: testing-by-betting simulation (the coin example)

```python
import numpy as np

def betting_wealth(flips, q=0.6):
    """flips: list of 0/1 (tails/heads). Bet towards alternative q."""
    wealth = [1.0]
    for x in flips:
        factor = 2*q if x == 1 else 2*(1-q)
        wealth.append(wealth[-1] * factor)
    return wealth

rng = np.random.default_rng(42)
flips = [1 if rng.random() < 0.6 else 0 for _ in range(30)]
wealth = betting_wealth(flips, q=0.6)

for t, (x, w) in enumerate(zip(flips, wealth[1:]), start=1):
    print(f"t={t:2d}  flip={'H' if x else 'T'}  W_t={w:.3f}")

alpha = 0.05
threshold = 1/alpha
print("Stop at t=12, wealth =", round(wealth[12], 3))
print("Continue to t=30, wealth =", round(wealth[30], 3))
print("Reject H0 at level", alpha, "?", any(w >= threshold for w in wealth))
```

7.7 The universal inference e-process I

Intuition. We want an e-process even when $\mathcal{P}$ is composite and no nontrivial test martingale family may exist. Idea: compare an *out-of-sample* alternative likelihood to an *in-sample* (i.e., overfit) null likelihood. Overfitting under the null only *helps* validity (it can only shrink the ratio), so this is safe.

Universal inference e-process (7.9)

$$E_0 = 1, \qquad E_t = \prod_{i=1}^{t} \frac{\hat{q}_{i-1}(X_i)}{\hat{p}_t(X_i)},$$

where $\hat{q}_{i-1} \in \mathcal{Q}$ may depend on $X_1, \ldots, X_{i-1}$ (updated online), and $\hat{p}_t \in \mathcal{P}$ is the MLE in $\mathcal{P}$ based on *all* of $X_1, \ldots, X_t$ (recomputed every step).

Proposition 7.18

E in (7.9) is an e-process for $\mathcal{P}$.

7.7 The universal inference e-process II

Proof idea: by definition of the MLE $\hat{p}_t$, for every fixed $\mathbb{P} \in \mathcal{P}$ with density p,

$$E_t \leq \prod_{i=1}^{t} \frac{\hat{q}_{i-1}(X_i)}{p(X_i)} =: (\text{test martingale for } \mathbb{P}),$$

since $\hat{p}_t$ maximizes the denominator (over $\mathcal{P}$) using the very same data, so replacing it with the true p can only make the ratio $\geq E_t$. The right side is a genuine test martingale for $\mathbb{P}$ (product of conditional likelihood ratios), which certifies E as an e-process.

7.7 The universal inference e-process III

Ratio interpretation: $\frac{\text{out-of-sample alternative likelihood}}{\text{in-sample null likelihood}}$. Each $\hat{q}_{i-1}$ is evaluated on a fresh point X_i outside the sample used to fit it; $\hat{p}_t$ is evaluated on all the data used to fit it. We "overfit" the null (safe) but not the alternative (would be unsafe). This avoids the complexity-adjustment headaches of generalized likelihood ratio tests, where thresholds must be enlarged to account for $\mathcal{P}, \mathcal{Q}$ complexity — here the threshold $1/\alpha$ never needs adjusting. A mixture variant, $E_t = \int_{\mathcal{Q}} \prod_i \frac{q(X_i)}{\hat{p}_t(X_i)} \, \mathrm{d}\nu(q)$, is also an e-process.

7.8 Sequential e-values I

Intuition. Imagine a chain of independent laboratories, lab $1, 2, 3, \ldots$, each producing one e-value E_t using only the results $E_1, \ldots, E_{t-1}$ reported by earlier labs (plus its own new data). Each lab's e-value is valid *conditionally* on everything reported before it. Multiplying all these e-values together builds a full e-process — evidence accumulates multiplicatively and validly across the whole chain.

Definition 7.19: sequential e-values

E-variables E_t, $t \in \mathbb{T}_+$, adapted to $\mathcal{F}$, are *sequential* if

$$\mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \leq 1 \quad \text{for all } t \in \mathbb{T}_+,\ \mathbb{P} \in \mathcal{P}.$$

(Default filtration: natural filtration of the E_t's themselves.)

Proposition 7.20

If $(E_t)_{t \in \mathbb{T}_+}$ are sequential e-variables for $\mathcal{P}$, then $M_t := \prod_{s=1}^{t} E_s$ (with $M_0 = 1$) is an e-process for $\mathcal{P}$.

7.8 Sequential e-values II

Proof: $\mathbb{E}^{\mathbb{P}}[M_t \mid \mathcal{F}_{t-1}] = M_{t-1}\,\mathbb{E}^{\mathbb{P}}[E_t \mid \mathcal{F}_{t-1}] \le M_{t-1}$, so M is a test supermartingale, hence an e-process.

7.8 Sequential e-values III

Betting on sequential e-values. Rather than betting *everything* on each E_t (fully multiplying), we can bet only a *fraction* $\lambda_t \in [0, 1]$ of wealth — convex-combining "do nothing" (factor 1) and "bet fully" (factor E_t).

Definition 7.21: e-process built on sequential e-values

(i)

$$M_t = \prod_{s=1}^{t} \big((1-\lambda_s) + \lambda_s E_s\big), \quad M_0 = 1, \tag{7.10}$$

where $\lambda_t \in [0,1]$ is predictable (a measurable function of $E_1, \ldots, E_{t-1}$).
(ii) The $\mathbb{Q}$-*log-optimal* choice solves $\lambda_t \in \arg\max_{\lambda \in [0,1]} \mathbb{E}^{\mathbb{Q}}[\log((1-\lambda) + \lambda E_t) \mid \mathcal{F}_{t-1}]$.
(iii) The *empirically adaptive* e-process (parameter $\gamma \in (0,1]$) instead solves, for $t \geq 2$,

$$\lambda_t \in \arg\max_{\lambda \in [0,\gamma]} \frac{1}{t-1} \sum_{s=1}^{t-1} \log\big((1-\lambda) + \lambda E_s\big), \qquad \lambda_1 = 0,$$

i.e., λ_t is chosen using only the *empirical* track record of past e-values, with no need to know a specific alternative $\mathbb{Q}$.

7.8 Sequential e-values V

Interpretation. The $\mathbb{Q}$-log-optimal choice maximizes "e-power" at each step assuming you know $\mathbb{Q}$; the empirically adaptive choice mimics this using only the empirical distribution of past e-values — a purely data-driven analogue with a default choice $\gamma = 1/2$. This construction is always a supermartingale (and a martingale if the E_t are exact), and M_τ is a valid e-variable at any stopping time τ.

Theorem 7.22

Suppose $(E_t)_{t\in\mathbb{N}}$ are iid under $\mathbb{Q}$ with $\mathbb{E}^{\mathbb{Q}}[\log E_1]$ finite. The empirically adaptive e-process M (with $\gamma = 1$) satisfies:

(i) *Asymptotic log-optimality:* $\lim_{t\to\infty} \frac{1}{t}(\log M_t - \log M'_t) \geq 0$ in $L^1(\mathbb{Q})$, where M' is the $\mathbb{Q}$-log-optimal e-process built on the same (E_t).

(ii) *Consistency:* if $\mathbb{E}^{\mathbb{Q}}[E_1] > 1$, then $M_t \to \infty$, $\mathbb{Q}$-almost surely.

Proof intuition: under the iid assumption, λ_t converges in probability to the fixed optimal λ^* that would be chosen with full knowledge of $\mathbb{Q}$; the two e-processes then share the same asymptotic growth rate by the law of large numbers.

Python: testing-by-betting with sequential e-values (empirically adaptive)

```python
import numpy as np

def empirically_adaptive(E, gamma=0.5):
    """E: array of sequential e-values E_1,...,E_T. Returns wealth M_0,...,M_T."""
    T = len(E)
    M = [1.0]
    lam = 0.0  # lambda_1 = 0
    log_growth_hist = []  # to average over E_1,...,E_{t-1}
    for t in range(T):
        M.append(M[-1] * ((1 - lam) + lam * E[t]))
        log_growth_hist.append(E[t])
        # choose lambda_{t+2} by grid search maximizing empirical mean growth
        grid = np.linspace(0, gamma, 200)
        avg_log = [np.mean(np.log((1 - l) + l * np.array(log_growth_hist))) for l in grid]
        lam = grid[int(np.argmax(avg_log))]
    return np.array(M)

rng = np.random.default_rng(0)
# Sequential e-values under an alternative where E[E_t] = 1.5 (favorable game)
E_vals = rng.exponential(scale=1.5, size=200)
M = empirically_adaptive(E_vals, gamma=0.5)
print("Final wealth after 200 rounds:", round(M[-1], 3))
print("Wealth path (every 20 steps):", np.round(M[::20], 3))
```

7.9 Time-mixture construction I

Intuition. Suppose at every sample size n you have a good e-variable $E^{(n)}$ (a "numeraire"), each individually valid, but the *sequence* $E^{(1)}, E^{(2)}, \dots$ is not necessarily an e-*process* (e.g., these could be entirely different statistics recomputed from scratch at each n, with no consistency across n). We can still combine them into a bona fide e-process by taking a weighted average across time — putting a probability distribution over "which n do I trust most," and never re-normalizing.

Time-mixture e-process (7.12)

Let w be any probability mass function on $\mathbb{N} = \{1, 2, \dots\}$. Define $M_0 = 1$ and

$$M_n = \sum_{j \le n} w(j)\, E^{(j)}.$$

Proposition 7.23

M defined above is an e-process for $\mathcal{P}$.

7.9 Time-mixture construction II

Proof: for any random time τ (not even required to be a stopping time!) and any $\mathbb{P} \in \mathcal{P}$,

$$\mathbb{E}^{\mathbb{P}}[M_\tau] = \mathbb{E}^{\mathbb{P}}\Big[\sum_{j\geq 1} w(j) E^{(j)} \sum_{n \geq j} \mathbb{1}_{\{\tau = n\}}\Big] \leq \sum_{j \geq 1} w(j)\, \mathbb{E}^{\mathbb{P}}[E^{(j)}] \leq 1,$$

using $\sum_{n\geq j} \mathbb{1}_{\{\tau=n\}} \leq 1$ and each $E^{(j)}$ being an e-variable.
Practical weight choice. $w(n) = c/(n(\log n)^2)$ (normalized so weights sum to 1) gives $\log M_n \geq \log E^{(n)} - \log n - 2\log\log n - O(1)$, so

$$\lim_{n\to\infty} \left(\tfrac{1}{n}\log M_n - \tfrac{1}{n}\log E^{(n)}\right) = 0 :$$

the time-mixed e-process matches the asymptotic growth rate of the underlying numeraire sequence, at essentially no cost.

Python: time-mixture construction

```python
import numpy as np

def time_mixture_weights(n_max):
    # w(n) = c / (n (log n)^2) for n>=2, normalized; w(1) fills remaining mass
    n = np.arange(2, n_max + 1)
    raw = 1.0 / (n * (np.log(n)) ** 2)
    w = np.zeros(n_max + 1)
    w[2:] = raw / raw.sum() * 0.99
    w[1] = 1 - w.sum()
    return w

def time_mixture_eprocess(E, w):
    # M_n = sum_{j<=n} w(j) E^(j); E[j-1] holds numeraire E^(j)
    return np.cumsum(w[1:len(E)+1] * E)

n_max = 500
w = time_mixture_weights(n_max)

# E^(n): fixed-n numeraire growing like exp(0.05 n) under the alternative
rng = np.random.default_rng(3)
logE = 0.05 * np.arange(1, n_max + 1) + rng.normal(0, 0.3, size=n_max)
E = np.exp(logE)

M = time_mixture_eprocess(E, w)
print("Growth rate of E^(n):", round(logE[-1] / n_max, 4))
print("Growth rate of M_n:  ", round(np.log(M[-1]) / n_max, 4))
```

7.10 A quantum-like puzzle for p-values I

The conceptual point. Whether a p-value is "valid" can depend not just on what actually happened, but on what *would have happened* in hypothetical alternate runs of the experiment that never occurred. E-processes never have this problem: validity only depends on the actual stopping time you actually used, because *every* stopping time is simultaneously valid.

Example 7.24 (paraphrased): Bob and Alice

Bob runs an A/B test intending to collect 10,000 observations. He looks once at 5000 points (but does not stop), then continues to 10,000 and reports $p = 0.04$. His boss Alice asks: "If you had seen > 4800 tails in the first 5000 tosses, what would you have done?" Bob admits he would have stopped early. Alice says this makes his reported $p = 0.04$ *invalid*: because Bob's actual stopping rule was implicitly data-dependent (sometimes 5000, sometimes 10,000, depending on a hypothetical world that never happened), the fixed-$n = 10{,}000$ p-value calculation no longer applies.

7.10 A quantum-like puzzle for p-values II

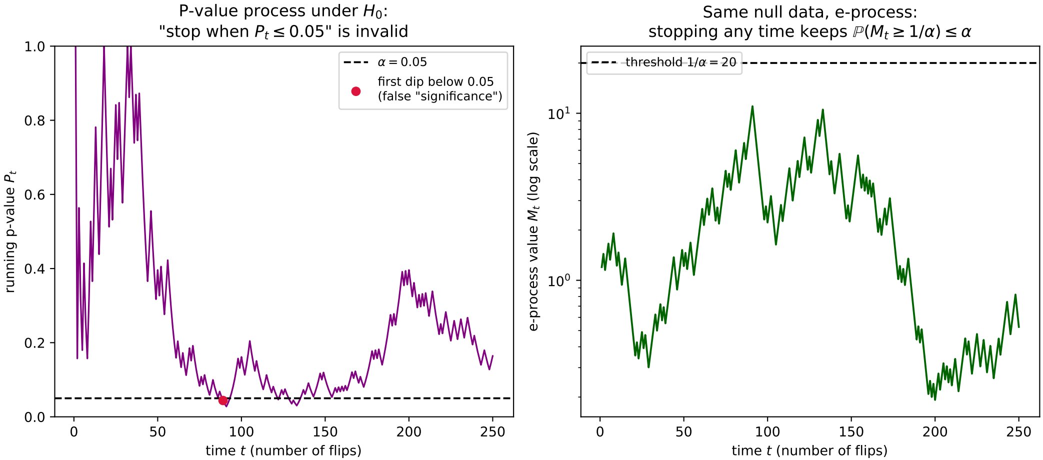
Why this is unsettling. The validity of Bob's p-value in the *actual, realized* world depends on his hypothetical answer about a world that did *not* occur. Merely being asked "what if" — and answering honestly — retroactively changes whether his real result is valid. This is impractical (nobody can enumerate all such counterfactuals) and conceptually troubling.

The e-process resolution

If Bob instead reports $E_{10,000}$ from a genuine e-*process* $(E_n)_{n\in\mathbb{N}}$ (not just a sequence of e-values), then **both** 5000 and $10{,}000$ are valid stopping times *simultaneously*, by construction of e-processes (Ville's inequality holds uniformly over all stopping times). It does not matter what Bob *would* have done in a hypothetical world at $n = 5000$ — whatever he would have reported there would also have been valid. There is no interference between hypothetical and actual worlds.

Takeaway. E-processes let you reason *forward only*, using just the data you actually observed and the stopping time you actually used — with no need to imagine what you would have done under alternate hypothetical stopping rules. This is a conceptual, not just technical, advantage of e-processes over p-values under optional stopping.

7.10 A quantum-like puzzle for p-values III



7.10 A quantum-like puzzle for p-values IV

Left: under a true null, the running p-value process can dip below 0.05 purely by chance at some intermediate time — "stop whenever $P_t \leq 0.05$" is an invalid procedure. **Right:** the corresponding e-process, built from the same data, stays comfortably under the threshold $1/\alpha$; stopping at *any* time (this one or any other) keeps $\mathbb{P}(M_t \geq 1/\alpha) \leq \alpha$ exactly as Ville's inequality guarantees.

7.11 When e-power misleads I

Intuition. "E-power" (expected log-growth, $\mathbb{E}^{\mathbb{Q}}[\log M_t]$) has been our main optimality criterion. This section shows two cautionary, somewhat pathological examples where e-power gives a *wrong impression* of an e-process's actual performance, because of subtleties with infinite expectations.

Two possible failure modes:

(a) an e-process can have *positive infinite* e-power at each fixed time, yet decline to 0 with probability 1 under $\mathbb{Q}$;

(b) an e-process can have *negative infinite* e-power at each fixed time, yet increase to ∞ with probability 1 under $\mathbb{Q}$.

Moral: extra caution is needed whenever infinities appear in e-power calculations.

7.11 When e-power misleads II

Example 7.25

Let $E_t = \exp(t^2 - Y_t)$, $t \in \mathbb{N}$, be e-variables for $\mathbb{P}$ ($\mathbb{E}^{\mathbb{P}}[E_t] \le 1$), where $Y_1, Y_2, \ldots$ are iid $\mathrm{Pareto}(1)$ under $\mathbb{Q}$ (i.e., $\mathbb{Q}(Y_1 > x) = 1/x$ for $x \ge 1$), independent under $\mathbb{P}$. Let $M_t = \prod_{s=1}^{t} E_s$ (an e-process by Proposition 7.20).
One shows, for any $x > 0$,

$$\mathbb{Q}(M_t \le x) = \mathbb{Q}\Big(\sum_{s=1}^{t} Y_s \ge \tfrac{t(t+1)(2t+1)}{6} - \log x\Big) \le t\,\mathbb{Q}\Big(Y_1 \ge \tfrac{(t+1)(2t+1)}{6} - \tfrac{\log x}{t}\Big) \to 0$$

as $t \to \infty$ (using a union bound and $\mathbb{Q}(Y_1 > x) = 1/x \to 0$). So $M_t \to \infty$ in $\mathbb{Q}$-probability — thresholding M_t rejects the null with probability approaching 1!
But: because $\mathrm{Pareto}(1)$ has infinite mean ($\mathbb{E}^{\mathbb{Q}}[Y_1] = \infty$),

$$\mathbb{E}^{\mathbb{Q}}[\log E_t] = t^2 - \mathbb{E}^{\mathbb{Q}}[Y_t] = t^2 - \infty = -\infty \quad \text{for every } t,$$

so $\mathbb{E}^{\mathbb{Q}}[\log M_t] = -\infty$ for every t — the exact same (uninformative) e-power as the useless *zero* process, despite M being a highly effective test in practice!

Chapter 7 summary I

- **Peeking at p-values** (7.1) is dangerous: stopping when $P_n \le \alpha$ can force $\mathbb{P}(P_\tau \le \alpha) = 1$ under the null ("sampling to a foregone conclusion").
- **Wald's SPRT** (7.2) shows sequential testing can be far more efficient than fixed-n testing, while foreshadowing the e-process machinery (the likelihood ratio process).
- **Test supermartingales and e-processes** (7.3) formalize "wealth that cannot systematically grow under the null": e-processes are the more general, more flexible notion, equivalent to being dominated by a test martingale family.
- **Ville's inequality** (7.4) is the time-uniform generalization of Markov's inequality: $\mathbb{P}(\exists t : M_t \ge 1/\alpha) \le \alpha$. It converts e-processes into level-α sequential tests, and every such test arises this way (Proposition 7.8).
- **Likelihood ratio processes** (7.5) are exactly log-optimal e-processes for simple nulls/alternatives, and asymptotically log-optimal (via plug-in estimation) for composite alternatives.
- **Testing by betting** (7.6) reframes all of this as a literal gambling game — wealth is evidence.
- **Universal inference** (7.7) builds e-processes for composite nulls via out-of-sample alternative-likelihood over in-sample null-likelihood (MLE) ratios.

Chapter 7 summary II

- **Sequential e-values** (7.8) can be multiplied, or fractionally bet on via $(1-\lambda)+\lambda E_t$, to build e-processes; the empirically adaptive choice of λ_t needs no knowledge of the alternative.
- **Time-mixture** (7.9) converts any sequence of per-n e-variables (numeraires) into a genuine e-process via a simple weighted sum, at negligible cost to asymptotic growth rate.
- **E-processes avoid hypothetical-world reasoning** (7.10): unlike p-values under optional stopping, their validity never depends on what you *would have* done in an alternate, unrealized run.
- **E-power has pathologies** (7.11): infinite expectations can make e-power wildly misleading; probabilistic convergence statements are an essential complement.